Steam Cylinder Lubrication

The current issue of *Lubrication* has an interesting and instructive article bearing on the subject of lubrication with which we not only cordially agree but as the matter is so clearly presented we take pleasure in making some brief abstracts. The author justly claims that it is far more difficult to ascertain whether a steam cylinder is efficiently and economically lubricated or not than it is to determine the same condition with bearings because the inside surface of the cylinder cannot be seen or felt while the engine is running. Wear in a cylinder cannot be usually detected except on examination when the cylinder head is removed. Cylinders and valves cannot be examined every day, so the tendency is to use an excess of oil in order to be sure of getting enough. A cylinder oil is required to lubricate every sliding surface which comes in contact with the steam. In reality this includes not only such surfaces as may actually be touched by the steam, but those moving parts which are heated by it or are under its pressure. The principal parts that must be lubricated are: The admission or controlling valve and valve rod, the cylinder walls, the piston and piston rings, the piston rod, and the exhaust valve if different from the admission valve. In addition the oil generally must lubricate the main stop valve and throttle. When various surfaces are usually under different pressure and temperature conditions in the same machines, and the lubricant must take care of their varying conditions equally satisfactorily, the slide valve, operating, as it does, with an excess pressure on one side, is exceedingly difficult to lubricate, as the tendency is to continuously scrape the oil away from the valve seat, giving a maximum opportunity for wear and leakage. This, of course, could be somewhat prevented if the edges of the valve and its seat could be chamfered, but while this procedure would assist in the lubrication it would result in poor steam economy and hence is unthinkable. To produce satisfactory lubrication of the slide valve, therefore, the oil film must be continually renewed. On account of this difficulty it has been found impractical to operate slide valves at a pressure much over 125 pounds per square inch.

Piston valves, being balanced as to steam pressure on the sliding surface, are not subjected to the tendency of squeezing the oil from between the bearing surfaces, but a seal must be maintained between the different parts of the valves to prevent leakage of steam and loss of power. The valve rods are usually comparatively small with little friction, but quite serious trouble may result in the running of the engine if they are not properly lubricated. When adjustment is very exact, any overheating may cause a change of alignment with a possible sticking of the valve and a waste of power.

While the nominal duty of the oil within the cylinder is to lubricate the cylinder walls, the proper lubrication of the piston rings and the piston rod is equally as important. Horizontal engines, on account of the weight of the piston, require greater care in the selection of a lubricant than do vertical cylinders, and it is considered the best practice in the horizontal type of engine to have the piston supported by a tail rod so that its weight is not carried by the cylinder wall directly. This allows the piston rings to function in producing a seal and not in carrying a load. Piston rings should be rounded on the edges so as to prevent the scraping off of the oil from the cylinder walls, and the rings also should be so designed as to float freely around the piston, thus taking up any misalignment due to unequal expansions or otherwise.

LUBRICATORS

There are two general types of lubricators used in steam cylinder lubrication: The hydrostatic type and the mechanical type. In some cases of direct application, small sight feed cups or similar devices are used.

The hydrostatic type of lubricator, working on the principle of forcing the oil into the lubricating line by means of a head of water produced by condensed steam, is somewhat erratic. Conditions may change the amount and temperature of the water and hence the head, and changes in temperature may vary the viscosity of the oil so that a constant and regular flow is not maintained. The amount of oil flowing into the lubricating line will be independent of the load or speed of the engine, and hence the lubricated parts will not receive the varying amounts of oil required to take care of the changing speed conditions.

In the mechanical type of lubricator, as it is generally geared directly to the engine, the oil is forced into the line at a rate which varies directly with the speed of the engine, thus giving a lubrication more commensurate with the requirements. When the engine stops, the lubricator also stops feeding, thus avoiding waste by flooding the engine with oil when it is not in operation. Most mechanical lubricators have sight feed cups so that the rate at which the oil is fed to the line can be ascertained at any time. In the multiple type of lubricator where one piece of apparatus feeds a number of lines, sight feeds with separate regulating devices should be installed in each line. The best types of lubricator will feed the same amount of oil regardless of temperature, viscosity or amount of oil in the receiver. Some types keep the feed pipes full by means of spring check valves, which also are vacuum proof. This causes oil to be fed the instant the engine is started. The mechanical lubricator can be used either with the direct system of application or with the atomization system, but the hydrostatic type is generally only used on the latter.

SUPERHEATED CONDITIONS

In engines with high superheat it has been the general practice to use heavy-bodied pure mineral oils, and as a result good lubrication has not always been obtained, and it has been the general opinion that such machines are very hard to lubricate. This use of a pure mineral oil is no doubt due to the fact that cylinder oils were too often used which were not properly refined or compounded with such fatty oils as would withstand the carbonizing effect at the high temperature of superheated steam. Unless very high superheat is used, steam at the exhaust will show some moisture, and as we must lubricate the cylinder on the exhaust stroke as well as on the expansion stroke, we must use a slightly compounded oil that will take care of slightly wet conditions. Even if the exhaust from the high pressure cylinder remains dry, that of the low pressure will be wet. This, of course, can be taken care of by using a different oil on the high and low pressure cylinders, but this is not necessary as oils are now made which will take care of both conditions satisfactorily.

DETERMINATION OF PROPER LUBRICATION

The real test of a cylinder oil is the condition of the rubbing surfaces after it has been used. In making an examination of a cylinder running on a certain oil, too hasty a conclusion as to its value should not be reached. It takes time to form a film surface on a cylinder wall, and if possible several weeks should elapse before the action of the oil is finally judged. A perfectly lubricated cylinder should be highly polished and have a glaze over its surface. It should have no rough or dull spots, and there should be no sign of rust if examined immediately after stopping the engine. The color should vary from a bright iron-white to a light brown or steel blue. If the cylinder head is removed immediately after stopping the engine there should be a film of oil all over the surface sufficiently thick to saturate three or four thicknesses of cigarette papers. Even when the engine has stood for several hours there should still be a film upon the cylinder walls, though not necessarily as thick as if examined immediately after stopping the engine. The stain on the paper should be brown. If it is black or has black particles or streaks, the cylinder or rings are probably wearing or the oil